LEARNINGS FROM THIS PROJECT

DATA PREPROCESSING			
#	Doubts / Issues / Errors	Clarification / Learning	Clarity
1	When checking null values at first, Total Charges column didn't had missing values, even it had blanks?	It was stored as string → So, no null values Once converted to numeric → caused NaN	Great learning
2	Why fill missing values in Total Charges with median?	Total Charges can have very high values (long-term customers) which are outliers. Median is safe → not affected by outliers.	
3	Why data types matter?	Wrong type → errors in EDA and modeling phases (e.g., object instead of numeric).	
4	Why don't we do one hot encoding in data preprocessing stage?	If we encoded the columns, then we won't be able to do EDA. Unable to work on Univariate and Bivariate analysis, as they are column based analysis	Great learning

UNIVARIATE & BIVARIATE ANALYSIS			
#	Doubts / Issues / Errors	Clarification / Learning	Clarity
1	Count vs Percentage in plots/tables	Count shows volume, but percentage gives real comparison (better for categorical data).	
2	KeyError in correlation	Correlation works only on numeric data, so target must be numeric (0/1) not (yes/no)	
3	Dataset balanced or imbalanced	This is important for modeling. So, we must check in EDA phase whether the dataset is balanced or imbalanced	Great learning
4	Creating new features (AvgMonthlySpend, NumServices)	This is feature engineering and this improves model.	Great learning

FEATURE SELECTION & MODEL CREATION			
#	Doubts / Issues / Errors	Clarification / Learning	Clarity
1	After encoding, why convert True/False → 0/1?	To keep all features consistent numeric format for ML models.	
2	Why multiple feature selection methods?	To compare → no single best method always	
3	Why use F1 score instaed of accuracy?	Because dataset is imbalanced → F1 balances precision & recall.	
4	Why accuracy misleading?	Model can predict majority class and still get high accuracy	
5	Is StandardScaler necessary?	Needed for distance-based models (KNN, SVM), not mandatory for trees	
6	Why RFE only used some models?	Some models (like RF, SVM) are too slow inside RFE. (RFE re-trains model repeatedly → computationally heavy)	
7	How to choose best k?	Compare performance → choose k with best overall model performance.	
8	How to know selected features?	Use selector.get_support() to extract feature names	Great learning
9	Feature selection vs model training confusion	Feature selection = choose inputs, model = learn patterns.	
10	How to handle imbalanced data when creating model?	Use SMOTE method on training data	Great learning
11	Is SMOTE = Oversampling? What it actually do?	Yes, but it's special type. Random Oversampling --> just duplicates data SMOTE --> creates new synthetic data (Instead of same rows duplicated, SMOTE creates new realistic artificial samples	Great learning
12	When to use SMOTE?	Model Training stage, because the goal is train model properly. So, when data imbalance is a problem we use SMOTE: - To help model learn minority class - To improve prediction	Great learning
13	Should SMOTE be used in feature selection?	No, Feature selection should use REAL DATA only, then only it can identify which input columns are truly important. If we use SMOTE, it will create new synthetic data, which may lead to pick wrong features	Great learning
14	Why use stratify when splitting train, test set?	Keeps same ratio of the target variable in train/test → important for imbalance. Without stratify, Train/Test split might become: Train → 90% No, 10% Yes Test → 50% No, 50% Yes ---> So, model learns wrong pattern With stratify, Train/Test both: 70% No 30% Yes ---> Balanced representation	Great learning
15	CV score vs Test score confusion	CV = training performance (How well will model perform in general?) Test = real-world performance (How well does it perform on new data?)	Great learning
16	Which model is best?	Choose based on Test F1 + stability, not just CV.	
17	Why performance drops in test?	Slight overfitting → normal behavior.	
18	We are doing feature selection methods with multiple models and comparing accuracy/F1 score, then why we are running again all models with gridsearch for model creation?	Feature selection stage ≠ Final model stage It finds good features, Models were NOT tuned - Rough best model GridSearch - Tune models - Final best model	Great learning

DEPLOYMENT			
#	Doubts / Issues / Errors	Clarification / Learning	Clarity
1	Why input must match original dataset?	Selector expects same structure as training	
2	Feature selection reduces features, so why full input?	Full input → selector picks required features internally Feature selection ≠ reduce user input It reduces what model USES internally User still gives normal input System handles everything behind the scenes	Great learning
3	Why not use inverse_transform for output?	Only needed for scaled outputs, not classification (0/1)	
4	Why encoding not reversed?	Because it's just label mapping, not transformation	Great learning
5	How real users give input?	Users give normal values → system does encoding internally	
6	Why use reindex in deployment?	To ensure same columns + same order as training	
7	Why use get_dummies in deployment?	To convert user input → model format	